

Nursery wound-cover test

Short note for Hannah Stewart and Coral Gardeners

Why test this

When coral fragments are cut, the new edge leaves exposed skeleton. Those spots can collect algae, sediment, and other fouling before tissue grows back. A small nursery test can tell us whether covering the wound improves healing enough to justify the added handling time.

Cover types to compare

Open wound No cover. Baseline comparison.	Concrete Low-cost mineral cover.	Apoxie Sculpt Epoxy treatment Adrian has used before.	Scripps material Confirm its use case, then include it as the new-material comparison.
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Design

- Use healthy nursery fragments from known parent colonies where possible.
- Make one small, repeatable wound on each fragment.
- Assign fragments at random to open wound, concrete, Apoxie Sculpt, or the Scripps material.
- Spread each cover type across parent colonies or genotypes, so the material effect is not mixed up with lineage.
- Spread treatments across the nursery instead of putting one cover type in one part of the array.
- Photograph each fragment the same way at every check.

What to measure

- Cover retention: is the material still attached?
- Tissue regrowth: how much live tissue has grown back across the wound?
- Fouling: algae, sediment, and other buildup on the wound or cover.
- Coral health: tissue loss, disease signs, bleaching, or death.
- New skeleton: visible scar filling or new skeleton on the cut surface.
- Handling cost: minutes per fragment and any added material cost.

Checks

T0	2 weeks	1 month	3 months	6 months
photo + wound area	attachment + fouling	early healing	healing + health	cost/value call

Scripps material check

The Scripps / Hybrid Reefs material belongs in this test if it can sit on a fresh coral wound. First, ask what the material is, how it cures or breaks down, whether it has touched living coral tissue before, and whether it is a cover, plug, tile, or settlement surface. If it is mainly a settlement surface, keep it as a separate nursery-surface test.

Nursery fragment wound-cover test

Compare simple wound covers and ask whether any is worth the extra handling.

A Check the Scripps / Hybrid Reefs material before adding it

LIKELY SCRIPPS LINK

CoralGuard / Coral Guard

Antifouling reef material.

Include it in this test only if it can cover wounds without irritating tissue.



MAY BE A SEPARATE TEST

SNAP-X and Brink

Scripps / UCSD settlement materials.

Likely better for larval settlement, substrate design, or fusion experiments.



ADD ONLY IF IT FITS

Can it touch a fresh wound?

Check adhesion, contact with fresh skeleton, curing or breakdown, and tissue irritation. If not, test it as a nursery surface instead.



B Nursery-fragment wound test

1 Group fragments

Group fragments by species, parent colony, genotype, size, and nursery neighborhood.

2 Same wound

Make one repeatable wound with the same target area. Photo immediately at T0.

3 Assign at random

Mix cover types within each group so parent colony does not explain the result.

Cover types

Open wound

Concrete patch

Apoxie Sculpt

Confirmed Scripps material

Healing measures

Wound-closure rate from repeated photos; percent fouling on exposed skeleton or cover; scar infill and new skeleton; tissue loss, disease signs, survival, and fragment growth.



Time and cost measures

Material cost; preparation time; underwater handling time; whether the cover falls off; rework or removal time; and fit with Coral Gardeners' normal nursery work.



Monitoring schedule



C Decision rule: keep a cover only if the benefit beats the work

Better healing

Faster closure, less fouling, more scar infill, better growth.

Extra work

Time, cost, retention failure, rework, and dive complexity.

No harm

No recession, disease, or mortality penalty.

Keep testing

Use in a larger test only if the result repeats.

Figure. The design keeps the test small, uses Coral Gardeners nursery fragments, and compares wound covers against an open-wound baseline.

Figure notes

- Panel A: check the Scripps / Hybrid Reefs material first: composition, cure or breakdown, and contact with living coral.
- Panel B: make one repeatable wound per fragment, assign treatments at random, and track the same measures through time.
- Panel C: keep a cover only if it heals faster, does not create health problems, and is fast enough for nursery work.

Questions for Hannah

- Which wound matters most for Coral Gardeners: a cut fragment end, or a side scrape like a donor-colony wound?
- What is the Scripps / Hybrid Reefs material: cover, plug, tile, or settlement surface?
- How many nursery fragments from known parent colonies can be used without disrupting production?
- Do the 2-week, 1-month, 3-month, and 6-month checks fit the normal nursery schedule?